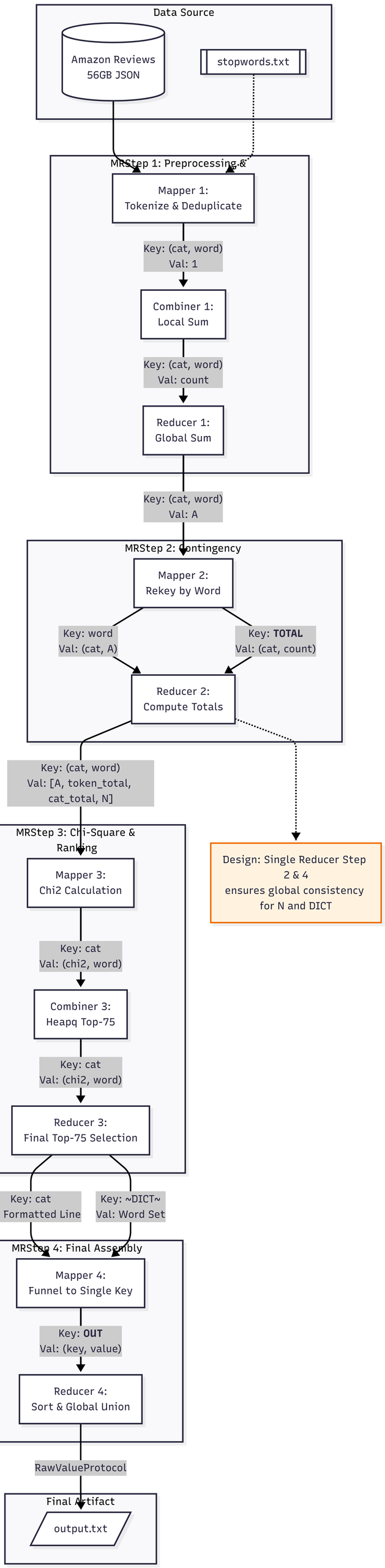
Data Source
Amazon Reviews
56GB JSON
stopwords.txt
MRStep 1: Preprocessing &
Mapper 1:
Tokenize & Deduplicate
Key: (cat, word)
Val: 1
Combiner 1:
Local Sum
Key: (cat, word)
Val: count
Reducer 1:
Global Sum
Key: (cat, word)
Val: A
MRStep 2: Contingency
Mapper 2:
Rekey by Word
Key: word
Val: (cat, A)
Key: TOTAL
Val: (cat, count)
Reducer 2:
Compute Totals
Key: (cat, word)
Val: [A, token_total,
cat_total, N]
Design: Single Reducer Step
2 & 4
ensures global consistency
for N and DICT
MRStep 3: Chi-Square &
Ranking
Mapper 3:
Chi2 Calculation
Key: cat
Val: (chi2, word)
Combiner 3:
Heapq Top-75
Key: cat
Val: (chi2, word)
Reducer 3:
Final Top-75 Selection
Key: cat
Val: Formatted Line
Key: ~DICT~
Val: Word Set
MRStep 4: Final Assembly
Mapper 4:
Funnel to Single Key
Key: OUT
Val: (key, value)
Reducer 4:
Sort & Global Union
RawValueProtocol
Final Artifact
output.txt